

Sara Chaudhari

s2chaudhari@ucsd.edu | +1 713 887 1369 | linkedin.com/in/sara-chaudhari-117791220

EDUCATION

University of California, San Diego , MS in Computer Science and Engineering	Sept 2024 – Jun 2026
• Relevant Coursework: Distributed Systems, Operating Systems, Advanced Algorithms, Computer Networks, ML Systems	GPA: 3.9/4.0
University of Mumbai , BE in Computer Engineering	Aug 2020 – Jun 2024
• Honors in Artificial Intelligence and Machine Learning	GPA: 4.0/4.0

EXPERIENCE

Software Engineer Intern, Genies	Sept 2025 – Dec 2025
• Built Python backend orchestration services for distributed AI workflows, implementing request queuing, connection pooling, and failure isolation to maintain service reliability under production load.	
• Optimized shared service throughput with Redis-backed response caching and batched async processing, improving capacity by 28% and reducing redundant compute under concurrent production traffic.	
• Designed shared configuration and retry utilities adopted across multiple backend services, standardizing failure handling and making service behavior more predictable across the platform.	
Software Engineer Intern, Dassault Systèmes Americas Corp.	June 2025 – Sept 2025
• Developed Java workflow execution components handling state transitions, job retries, and lifecycle management across 2,000+ pipeline artifacts, preventing upstream dependency failures from propagating downstream.	
• Containerized Python workloads using Docker and Kubernetes with Terraform-managed infrastructure, adding parallel execution and retry logic to handle dependency failures across batch workflows.	
• Automated Jenkins CI/CD validation pipelines using Groovy and Spock, covering regression scenarios and reducing production defects by 20% through systematic pre-deployment testing.	
Software Engineer, UpSolve Solutions	Apr 2023 – Jul 2024
• Engineered C++ services on Linux to process streaming user events and execute pre-trained models for real-time scoring, reducing end-to-end latency by 30% across 5,000+ daily requests.	
• Designed async pipeline with Redis queues and AWS SQS using producer-consumer batching, decoupling compute from request handling and improving throughput by 22% during peak load periods.	
• Instrumented monitoring pipeline comparing live model outputs against offline benchmarks to detect feature inconsistencies early, publishing service health telemetry and reducing debugging cycles.	
Software Engineer, Garg Group	Sept 2022 – Apr 2023
• Architected automated data ingestion and feature generation pipelines, orchestrating preprocessing stages and scheduled inference runs and reducing manual pipeline overhead by 40%.	
• Processed 100K+ supply chain time-series records using Spark on AWS EMR, generating time-based features and reducing end-to-end pipeline latency by 33%.	
• Orchestrated Airflow-scheduled scoring and evaluation workflows, improving anomaly detection accuracy by 15% over rule-based baselines and enabling automated alerting for operations teams.	

RESEARCH EXPERIENCE

Graduate Researcher, UC San Diego	Mar 2025 – Sept 2025
• Investigated causal discovery from observational data under distribution shifts and hidden confounding, implementing constraint-based and score-based methods across 40+ controlled experiments.	
• Evaluated causal inference workflows incorporating nonlinear models and missing data handling, achieving 18% improvement in structural recovery accuracy over baseline methods.	
• Constructed reproducible experimentation pipelines on Linux-based Kubernetes clusters, enabling consistent cross-environment evaluation using structural Hamming distance and intervention accuracy metrics.	

KEY PROJECTS

Distributed Search Engine	
• Implemented distributed search engine with multithreaded ingestion, shard-based indexing, and parallel query execution, achieving sub-5-second latency across 50GB+ of data.	
• Benchmarked index lookup and query scheduling under concurrent load, improving search throughput and maintaining consistent response times across varied query distributions.	
Service Availability Monitor	
• Built fleet service health monitor in Python, polling availability and latency metrics from distributed endpoints and persisting device deployment state in a Postgres-backed store.	
• Implemented Redis pub/sub alerting pipeline with threshold-based anomaly detection, validating behavior across 15+ simulated failure scenarios and reducing mean time to identify degraded endpoints.	

PUBLICATIONS

• Genome Sequence Analysis and Drug – Target Interaction Prediction using Deep Learning (Link)	
Journal: Springer Innovative Computing and Communications, LNNS Series (Vol. 1038)	
• Two-Pass Pipeline for Automated Medical Infographic Generation Using Text Summarization	
Presented in SmartDataCom: International Conference on Smart Data Processing, Communication and Networking (2024)	

SKILLS

Languages: C++, Python, Java, TypeScript, SQL
Systems & Backend: Linux, Multithreading, Async Architecture, Concurrency, Distributed Task Orchestration
Cloud & Infra: AWS, Terraform, Docker, Kubernetes, ECS, S3, Postgres, Redis, SQS, CI/CD
Observability: Service Telemetry, Production Debugging, Benchmarking, Fault-Tolerant Pipelines
Developer Tools: Cursor, Claude Code, GitHub Copilot, Jenkins, Git, Airflow, Spark